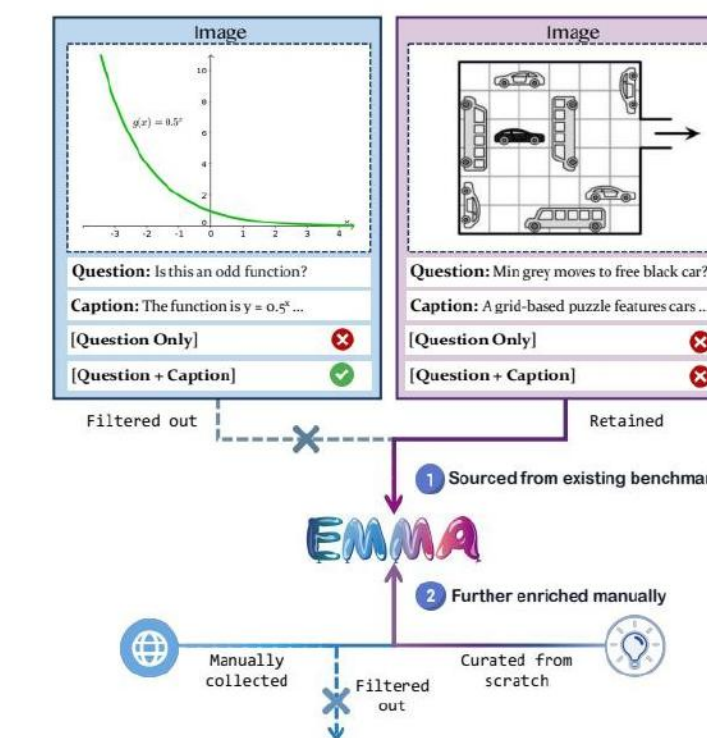


# Can MLLMs Reason in Multimodality? EMMA Benchmark

UESTC; SYSU; UW; Microsoft; CUHK

## Problem Overview

Setting and Motivation



## Force Direction

Fig. 1: Force direction

Curved path with v arrows: where is net acceleration?  
Read diagram; relate curvature to centripetal a and dv

Physics: Visual Decomposition Simulation

Question:

An object travels along a path shown above, with changing velocity as indicated by vectors A and B. Which vector best represents the net acceleration of the object from time  $t_A$  to  $t_B$ ?

Ground Truth:

- Identify the direction of acceleration, same as the direction of the change in velocity.
- Because:  $\Delta \vec{v} = \vec{B} - (-\vec{A})$ , the direction  $\Delta \vec{v}$  of can be determined graphically.
- Placing the B vector "tip-to-tail" with the -A vector gives a direction for  $\Delta \vec{v}$  (and therefore, a) to the left, and therefore, a) to the left.

GP2-40

- Velocity at  $t_A$  ( $\vec{v}_A$ ):  $\vec{v}_A$  points upward (+y) and to the right (+x).
- Velocity at  $t_B$  ( $\vec{v}_B$ ):  $\vec{v}_B$  points upward (+y) and to the left (-x).
- Acceleration:  $\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_B - \vec{v}_A}{\Delta t} = \frac{\vec{v}_B + (-\vec{v}_A)}{\Delta t}$ . Here,  $-\vec{v}_A$  points downward (-y) and to the left (-x).
- Net Direction: Horizontal components (+x, -x) partially cancel. Vertical components (+y, -y) combine, resulting in a downward (-y) vector.
- Conclusion: The net acceleration points downward (-y), corresponding to option b.

## Key Definitions

Scope: math, physics, chem, coding; MCQ + free

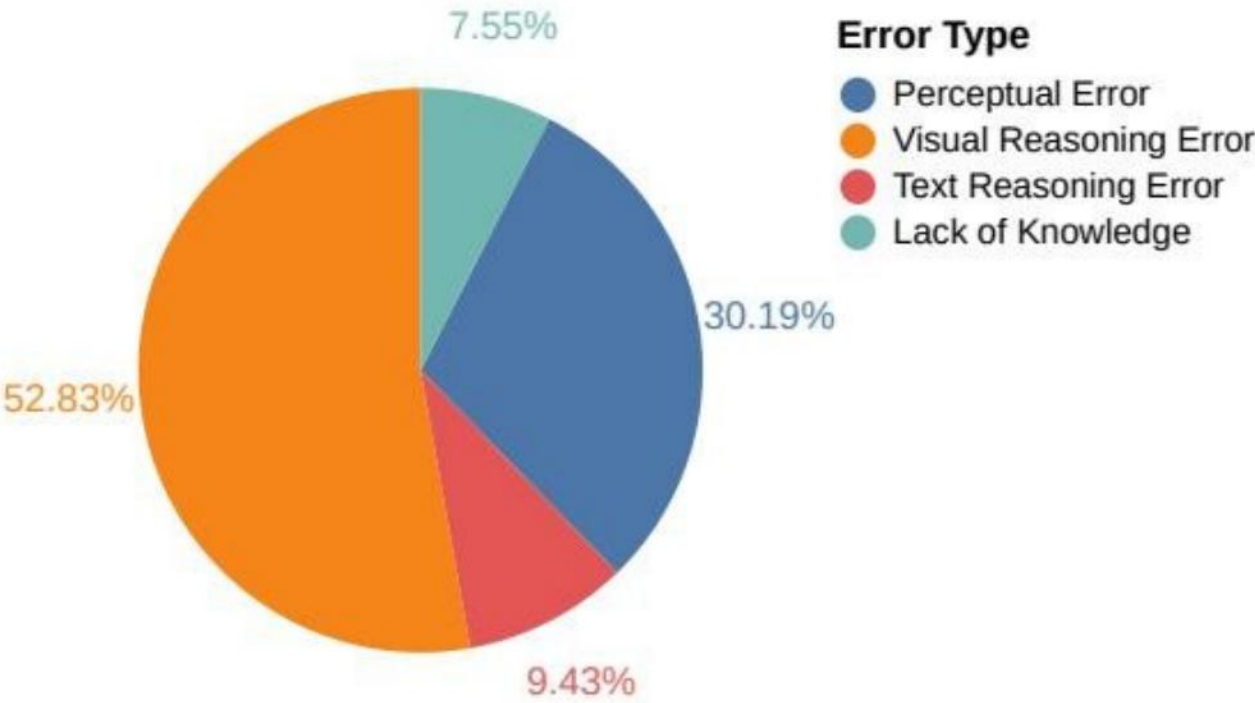
Organic multimodal: vision + multi-step reasoning

Objective: accuracy vs. human; per-skill diagnostics



# Why Methods Fail

Visual Reasoning Dominates  
Over half of errors visual: spatial and transformations  
CoT helps at times, scaling or tuning won't integrate



# Prompts & Compute

Test-time scaling: sample N=1,2,4,8,16; rerank/select

Standard splits + unified prompting: MCQ+open; dir/CoT

Gains plateau; big gap on spatial and multi-hop tasks

# Curation & Labels

Subject-specific data + checks; plausible distractors

Two-step curation: captions + rules purge shortcuts

Balanced stats; pressures visual-to-symbolic reasoning

# EMMA's Question: Can MLLMs Reason Multimodally?

UESTC; Sun Yat-sen Univ.; UW; Microsoft; CUHK



# EMMA Metrics Gap

Metrics and Human Gap

EMMA/mini: overall, per-category accuracy; baselines

Path, 3D, multi-hop, pattern: large human gap

Identical prompts; strong distractors expose reasoning

|                                    | EMMA-mini        |              | EMMA       |            |            |            |                  |
|------------------------------------|------------------|--------------|------------|------------|------------|------------|------------------|
|                                    | Overall<br>(100) | Path<br>(13) | 3D<br>(37) | MH<br>(33) | VD<br>(47) | GR<br>(26) | Overall<br>(156) |
| Random choice                      | 23.00            | 38.46        | 21.62      | 27.27      | 31.91      | 11.54      | 25.64            |
| Human Expert                       | 64.50            | -            | -          | -          | -          | -          | -                |
| Direct Claude 3.5 Sonnet           | 34.00            | 30.77        | 37.84      | 36.36      | 31.91      | 30.77      | 33.97            |
| Direct Gemini 2.0 Flash            | 40.00            | 38.46        | 29.73      | 42.42      | 38.30      | 46.15      | 38.46            |
| Direct GPT-4o                      | 38.00            | 30.77        | 40.54      | 33.33      | 36.17      | 50.00      | 38.46            |
| Direct Qwen2-VL-72B-Instruct       | 40.00            | 30.77        | 48.65      | 45.45      | 42.55      | 34.62      | 42.31            |
| Direct LLaVA-Onevision-72B         | 32.00            | 23.08        | 27.03      | 39.39      | 48.94      | 26.92      | 35.90            |
| Direct InternVL2-Llama3-76B        | 22.00            | 15.38        | 32.43      | 21.21      | 6.45       | 23.08      | 22.44            |
| Direct InternVL2.5-78B             | 40.00            | 38.46        | 43.24      | 42.42      | 38.30      | 26.92      | 38.46            |
| CoT Claude 3.5 Sonnet              | 38.00            | 53.85        | 37.84      | 36.36      | 42.55      | 42.31      | 41.03            |
| CoT Gemini 2.0 Flash               | 41.00            | 30.77        | 29.73      | 36.36      | 44.68      | 46.15      | 38.46            |
| CoT GPT-4o                         | 44.00            | 69.23        | 35.14      | 39.39      | 44.68      | 46.15      | 43.59            |
| CoT Gemini 2.0 Flash Thinking-1219 | 57.00            | 61.54        | 43.24      | 57.58      | 61.70      | 61.54      | 56.41            |
| CoT Gemini 2.0 Flash Thinking-0121 | 63.00            | 61.54        | 54.05      | 60.61      | 57.45      | 73.08      | 60.26            |
| CoT OpenAI o1                      | 49.00            | -            | -          | -          | -          | -          | -                |
| CoT Qwen2-VL-72B-Instruct          | 34.00            | 15.38        | 35.14      | 33.33      | 34.04      | 46.15      | 34.62            |
| CoT LLaVA-Onevision-72B            | 26.00            | 15.38        | 10.81      | 18.18      | 6.38       | 15.38      | 12.18            |
| CoT InternVL2-Llama3-76B           | 33.00            | 53.85        | 27.03      | 21.21      | 12.90      | 38.46      | 32.05            |
| CoT InternVL2.5-78B                | 36.00            | 46.15        | 35.14      | 30.30      | 46.81      | 42.31      | 39.74            |

# Filtering Rules

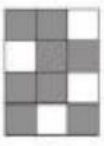
Enforce Visual Reasoning

Not solvable from captions; plausible visual foils

Multi-step: rotation, symmetry; minimal domain facts

▶ Rotation

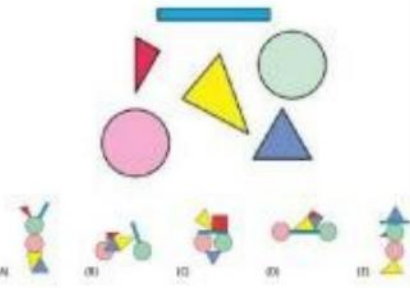
Image:



Question: Which of the rectangles A to E can be covered by the pattern on the right-hand side in such a way that the result is a totally black rectangle?

▶ Translation

Image:



Question: Which of the following pictures can you NOT do with all the pieces beside?

▶ Flip

Image:



Question: Mr Hofer has drawn a picture of flowers on the inside of a display window (large picture). What do these flowers look like when you look at the picture from the outside?

# Negative Examples

Wedge rotations: miss global rule; chase local cues

Cube net: opposite faces/sums; many choose wrong folds

Tiny cases need precise, multi-step visual integration

Gaps and Gains at a Glance

MLLMs trail humans on EMMA and EMMA-mini

Largest gaps: 3D/spatial sim, planning, multi-hop

Test-time scaling helps; the human gap persists

One-pass prompting strong; fails on visual relations

Bottleneck: visual reasoning, not perception

|                                    | EMMA-mini        |             | EMMA      |             |                 |             |                   |
|------------------------------------|------------------|-------------|-----------|-------------|-----------------|-------------|-------------------|
|                                    | Overall<br>(100) | SR<br>(474) | GR<br>(9) | RS<br>(132) | RS-pro<br>(105) | KS<br>(456) | Overall<br>(1176) |
| Random choice                      | 27.00            | 24.47       | 33.33     | 27.27       | 35.24           | 0.44        | 16.50             |
| Human Expert                       | 86.00            | -           | -         | -           | -               | -           | -                 |
| Direct Claude 3.5 Sonnet           | 44.00            | 66.88       | 22.22     | 55.30       | 55.24           | 6.80        | 40.90             |
| Direct Gemini 2.0 Flash            | 36.00            | 54.01       | 11.11     | 53.79       | 58.10           | 8.33        | 36.31             |
| Direct GPT-4o                      | 33.00            | 47.05       | 11.11     | 51.52       | 42.86           | 8.33        | 31.89             |
| Direct Qwen2-VL-72B-Instruct       | 34.00            | 45.99       | 33.33     | 48.48       | 45.71           | 9.65        | 32.06             |
| Direct LLaVA-Onevision-72B         | 24.00            | 38.19       | 33.33     | 39.39       | 26.67           | 7.24        | 25.26             |
| Direct InternVL2-Llama3-76B        | 21.00            | 37.34       | 22.22     | 31.45       | 24.76           | 8.55        | 24.06             |
| Direct InternVL2.5-78B             | 38.00            | 55.06       | 33.33     | 47.73       | 43.81           | 8.99        | 35.20             |
| CoT Claude 3.5 Sonnet              | 41.00            | 57.17       | 33.33     | 58.33       | 58.10           | 15.57       | 41.07             |
| CoT Gemini 2.0 Flash               | 36.00            | 22.15       | 33.33     | 50.00       | 59.05           | 11.84       | 24.66             |
| CoT GPT-4o                         | 35.00            | 42.41       | 33.33     | 51.52       | 45.71           | 16.67       | 33.67             |
| CoT Gemini 2.0 Flash Thinking-1219 | 41.00            | 48.31       | 33.33     | 45.45       | 69.52           | 17.76       | 37.93             |
| CoT Gemini 2.0 Flash Thinking-0121 | 47.00            | 53.16       | 66.67     | 48.48       | 58.10           | 23.25       | 41.58             |
| CoT OpenAI o1                      | 40.00            | -           | -         | -           | -               | -           | -                 |
| CoT Qwen2-VL-72B-Instruct          | 32.00            | 33.33       | 11.11     | 37.12       | 42.86           | 7.89        | 24.57             |
| CoT LLaVA-Onevision-72B            | 23.00            | 33.76       | 0.00      | 37.88       | 20.95           | 7.24        | 22.53             |
| CoT InternVL2-Llama3-76B           | 21.00            | 29.11       | 22.22     | 30.65       | 22.86           | 6.58        | 19.73             |
| CoT InternVL2.5-78B                | 24.00            | 37.13       | 33.33     | 37.12       | 33.33           | 13.16       | 27.47             |